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AURORA: Paperless Academic Defense Workflow System for Research, Capstone, Thesis & Dissertation Papers at Partido State University

Gene Vincent D. Mañago
BSIT 4C Proponent / Lead Dev
Dept. of Computational Sciences
Partido State University, Goa

Carol Ann B. Castillo
BSIT 4C Proponent / Analyst
Dept. of Computational Sciences
Partido State University, Goa

Rona Mae S. Buenafe
BSIT 4C Proponent / QA Lead
Dept. of Computational Sciences
Partido State University, Goa

Lara Mae B. Encinas
BSIT 4C Proponent / UI-UX
Dept. of Computational Sciences
Partido State University, Goa

Leomer Bares
BSIT 4C Proponent / Database
Dept. of Computational Sciences
Partido State University, Goa

Froiland B. Peñalosa
Capstone Adviser & Coordinator
Dept. of Computational Sciences
Partido State University, Goa

Abstract

The academic defense process is a cornerstone of higher education quality assurance, validating student competencies in research, capstone projects, theses, and dissertations [1]. However, at Partido State University – Goa Campus, existing defense workflows across its five constituent colleges remain predominantly paper-based and manually coordinated. Students print multiple bulky manuscript drafts per stage; advisers and panelists record handwritten marginalia across separate copies; coordinators manually verify schedules; and evaluators compute rubric scores on physical sheets, inducing cognitive split-attention [2], prolonged turnaround times, high printing costs [3, 4], and fragmented revision tracking. This paper presents AURORA (Paperless Academic Defense Workflow System for Research, Capstone, Thesis & Dissertation Papers at Partido State University), a cloud-native web platform engineered to digitize and unify the end-to-end academic defense lifecycle. Guided by the Agile Software Development Life Cycle (SDLC) and Design Science Research [7], AURORA implements a three-tier architecture built with Next.js (App Router), React, TypeScript, Tailwind CSS, and Supabase (PostgreSQL with Row-Level Security across 26 normalized tables). The platform integrates nine core functional modules: (1) multi-role authentication and Role-Based Access Control (RBAC) [8]; (2) multi-college configurable defense workflows; (3) SHA-256 versioned PDF manuscript submission; (4) an interactive split-screen PDF.js annotation workspace with normalized coordinate pinning; (5) automated weighted rubric evaluation and real-time score consolidation [1, 9]; (6) unified revision tracking and approval verification [11]; (7) conflict-free defense scheduling with an adviser conflict-of-interest guard; (8) real-time notifications and ISO/IEC 27001 audit trails [14]; and (9) a 3-2-1 Disaster Recovery and Self-Healing Maintenance Console. Empirical evaluation with N = 27 institutional stakeholders using a validated 10-item System Usability Scale (SUS) [12] and 27-item UTAUT acceptance survey [6] (Cronbach's α = 0.946) yielded a composite SUS score of 91.8 / 100 (Grade A — Excellent) [13] and a grand UTAUT mean of 4.86 / 5.00 (Highly Acceptable), while reducing manuscript endorsement turnaround time by 85.9% (from 18.4 days to 2.6 days).

Keywords: *Academic Defense Workflow, Capstone, Digital Rubric, Higher Education, Manuscript Management, Paperless System, Partido State University, PDF Annotation, Role-Based Access Control, System Usability Scale, Thesis, UTAUT, Web-Based Platform.*

1 Introduction

The academic defense process serves as a foundational quality assurance mechanism in higher education, validating student competencies across undergraduate research, capstone projects, master's theses, and doctoral dissertations [1]. Despite global strides in higher education digital transformation [5], requirements-gathering interviews conducted with Research Coordinators across all five constituent colleges of Partido State University (ParSU) – Goa Campus—namely the College of Engineering and Computational Sciences (CECS), College of Education (COE), College of Arts and Humanities (CAH), College of Science (COS), and College of Business and Management (CBM)—revealed that defense workflows remain heavily reliant on physical paper routing and fragmented digital utilities.

Under the existing process, student teams print 3 to 5 copies of 80-to-200-page manuscripts per defense stage, incurring heavy financial burdens and paper consumption [3, 4]. Advisers and panel evaluators write marginalia across separate physical printouts, forcing students to reconcile feedback from disconnected sources and inducing split-attention cognitive overload [2]. Furthermore, manual rubric scoring on paper sheets delays grade consolidation [1, 10], while advisers lack side-by-side version tracking to verify whether panel revisions were properly addressed [11].

The main objective of this study is to design, develop, and evaluate AURORA, a web-based paperless academic defense workflow system for research, capstone, thesis, and dissertation papers at Partido State University – Goa Campus. Specifically, it fulfills four objectives: (Obj. 1) gather requirements by identifying functional and non-functional needs for paperless defense workflows across research, capstone, and dissertation papers; (Obj. 2) design the system architecture, 26-table database schema, user interface, and workflows supporting manuscript submission, annotation, rubric-based evaluation, and revision tracking; (Obj. 3) develop the academic defense workflow system by integrating role-based access control, annotation workspace, rubric-based assessment, and automated scoring modules; and (Obj. 4) evaluate the system's usability, efficiency, and acceptance using standardized instruments such as the System Usability Scale (SUS) [12] and UTAUT-based acceptance surveys [6].

2 Related Work & Theoretical Framework

AURORA synthesizes five theoretical foundations: (1) Sweller's Cognitive Load Theory and Split-Attention Effect [2], which motivates integrating the PDF manuscript viewer, coordinate

annotations, and rubric drawer into a single split-screen workspace; (2) Ferraiolo and Kuhn's Role-Based Access Control (RBAC) [8], governing strict privilege isolation across Students, Advisers, Panelists, Coordinators, Deans, and System Administrators; (3) Spady's Outcome-Based Education (OBE) [9] and Erradi et al.'s automated rubric assessment [1, 10]; (4) Hevner et al.'s Design Science Research (DSR) [7] combined with ISO/IEC 27001:2022 [14] and Republic Act No. 8792 [15] for tamper-evident audit trails and digital signatures; and (5) Brooke's SUS [12, 13] and Venkatesh et al.'s UTAUT [6] for empirical usability and acceptance evaluation.

3 Methodology

3.1 Agile SDLC & Multi-College PIECES Analysis

AURORA was developed using the Agile Software Development Life Cycle (SDLC) from August 2025 to November 2026. Structured interviews with seven Research Coordinators across all five colleges at ParSU – Goa Campus were analyzed using the PIECES Framework (Performance, Information, Economics, Control/Security, Efficiency, and Service), establishing support for both 4-stage Capstone Defenses (CECS) and 3-stage Thesis/Research Defenses (COE, CAH, COS, CBM).

3.2 Three-Tier Architecture & 26-Table Database Schema

As summarized in Table 1, AURORA employs a Three-Tier Cloud Architecture backed by 26 normalized PostgreSQL tables (campuses, colleges, departments, programs, profiles, roles, user_roles, workflow_templates, defense_stages, projects, project_members, documents, document_versions, annotations, annotation_replies, rubric_templates, rubric_criteria, evaluations, evaluation_scores, defense_schedules, defense_panels, revision_items, notifications, audit_logs, etc.) protected by Row-Level Security (RLS).

Table 1. AURORA Three-Tier Architecture & Functional Stack

Tier	Core Stack	Modules & Governance Role
1. Presentation	React 19, PDF.js	60/40 split-screen PDF & rubric workspace
2. Application	Next.js, TS	9 modules; RBAC; RA 8792 step-up auth
3. Data Tier	Supabase PG 15	26-table RLS; SHA-256 versions; 3-2-1 DR

3.3 Coordinate Annotations, Auto-Rubrics & Evaluation

PDF highlights on page p are normalized against canvas dimensions (Cw, Ch) into floating-point percentages (x%, y%, w%, h%) ∈ [0, 100], preventing positional drift across zoom levels. During defense grading, the rubric engine computes weighted scores $Sw = \Sigma(ri \cdot wi / 100)$ and locks evaluations under RA 8792 via step-up password re-authentication and atomic serial generation (AURORA-YYYY-NNNNNN). For Objective 4, the 10-item SUS [12] and 27-item UTAUT survey [6] (excluding redundant items PE2 and UB2 per expert review; Good & Scates validity = 4.89/5.00; Cronbach's $\alpha = 0.946$) was administered to N = 27 ParSU stakeholders.

4 Results and Discussion

4.1 Workflow Efficiency & Multi-College Verification

Replacing physical manuscript routing with AURORA's cloud workflow reduced average adviser endorsement turnaround time from 18.4 days to 2.6 days—an 85.9% reduction (Table 2)—while eliminating printing expenses (100% paperless) and room/panelist scheduling conflicts (0.0% clash rate across 42 test cases). Furthermore, the integrated Maintenance & Disaster Recovery Console (/admin/system-health) provides 6 live subsystem probes, self-healing pointer repair, and SHA-256 verified JSON cold backups.

Table 2. Operational Comparison: Paper Workflow vs. AURORA

Operational Metric	Paper Process	AURORA Platform
Endorsement Turnaround	18.4 days	2.6 days (−85.9%)
Manuscript Pages Printed	1,800+ pages	0 pages (Paperless)
Rubric Consolidation	45–60 mins	< 1.0 sec (Instant)
Room / Panel Clashes	14.2% rate	0.0% (Hard Guard)
DR Backup RTO / RPO	None (Manual)	≤ 4h RTO / WAL PITR

4.2 Objective 4: Empirical SUS & UTAUT Results

Table 3 summarizes the empirical evaluation (N = 27). AURORA achieved a composite SUS score of 91.8 / 100 (normalized mean $\mu = 4.84$, SD = 0.28), earning a Grade A (Excellent) usability rating [13]. Across the six UTAUT constructs, Behavioral Intention rated highest ($\mu = 4.90$, SD = 0.20), followed by Performance Expectancy ($\mu = 4.88$, SD = 0.22), Use Behavior ($\mu = 4.87$, SD = 0.24), Effort Expectancy ($\mu = 4.86$, SD = 0.25), Facilitating Conditions ($\mu = 4.85$, SD = 0.26), and Social Influence ($\mu = 4.81$, SD = 0.30), yielding a Composite Grand Mean of 4.86 / 5.00 (Highly Acceptable).

Table 3. Empirical SUS & UTAUT Evaluation Summary (N = 27)

Evaluation Construct	Mean	SD	Verbal Rating
Part I: SUS (10 Items)	4.84	0.28	Grade A (91.8/100)
PE: Performance Expectancy	4.88	0.22	Highly Acceptable
EE: Effort Expectancy	4.86	0.25	Highly Acceptable
SI: Social Influence	4.81	0.30	Highly Acceptable
FC: Facilitating Conditions	4.85	0.26	Highly Acceptable
BI: Behavioral Intention	4.90	0.20	Highly Acceptable
UB: Use Behavior	4.87	0.24	Highly Acceptable
Composite Grand Mean	**4.86**	**0.25**	**Highly Acceptable**

5 Conclusion and Recommendations

AURORA successfully modernizes the academic defense lifecycle for research, capstone, thesis, and dissertation papers at Partido State University – Goa Campus. By unifying configurable multi-college workflows, coordinate-pinned PDF annotations, automated weighted rubric evaluation, RA 8792 digital signatures, and 3-2-1 disaster recovery, the system achieves an 85.9% reduction in review turnaround time, a 91.8/100 SUS score, and a 4.86/5.00 UTAUT acceptability rating.

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